

Lab 3: Architectural Pattern Justification

Digital Art Commission & Watermarking Portal

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Architecture Selection

We chose Layered Architecture for the Digital Art Commission & Watermarking Portal System.

The system has three layers:

the Presentation Layer (Web Portal UI), the Business Layer (Commission Manager, Watermark Service and Payment Service), and the Data Layer (database and private S3 files). The Web Portal UI connects to the Commission Manager through IOrder. The Commission Manager uses IWatermark to watermark drafts, IPayment to handle payments, and IStorage to store and access order data and files.

Reasons for the Choice

1. Clear separation of concerns: The system separates the UI, business logic and data storage into different layers. This makes development and testing easier. Components can be changed without affecting others, as long as their interfaces remain the same.

2. Suitable for a small system: The commission process follows a fixed sequence, from submitting a brief to unlocking the final file. The Commission Manager handles this process in one place. Microservices would make the system more complex than necessary, while a simple client-server design could mix all the business logic together.

Security Advantage

The Web Portal UI cannot access the database or files directly. All requests go through the Business Layer, which checks permissions and payment status. Only watermarked drafts are stored in the draft area. The original high-quality file is shared through a temporary link that expires after 60 minutes, and only after payment is confirmed. This prevents users from accessing the original file before final payment (FR-001, NFR-001).

Performance Benefit

Because the layers are separate, the heavy work (applying watermarks to images) is isolated in the Watermark Service, and the Business Layer can be scaled or cached independently of the UI. Downloads of final files go straight to storage through the signed URL, so large files do not pass through the application logic and the Commission Manager stays responsive. The main cost is a small overhead from layer-to-layer calls, which is acceptable at the expected load.